

A Unified Multiphysics Model of Pacinian Mechanotransduction Coupling Solid Mechanics and Neural Propagation

Modelling of Multiphysics Phenomena – Project Presentation

Edoardo Meconi

Riccardo Mambrini

University of Pisa

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Outline

- 1 Introduction
- 2 Model Geometry
- 3 Mechanics
- 4 Electric Model
- 5 Study
- 6 Conclusion

Introduction: Problem Definition & Research Context

- **Motivation & Ultimate Goal:**

- To demonstrate the feasibility of a **fully unified computational model** entirely within COMSOL , overcoming traditional multi-software approaches[1].

- **Our Unified Solution:**

- Implementation of an integrated **Unified COMSOL Framework**.
- Direct coupling between mechanical deformation and electro-ionic gating equations.

- **Research Objectives:**

- 1 To validate this seamless transduction model within a single software environment.
- 2 To demonstrate the emergence of FA behavior from pure structural-viscoelastic principles.
- 3 To pave the way for the future design of **bio-inspired tactile sensors** for robotics and prosthetics.



Figure: COMSOL Software Logo

Introduction: Anatomy of Pacinian Corpuscle

Target: Understanding the Pacinian Corpuscle's role as a high-frequency vibration sensor.

- **Anatomy:**

- Encapsulated structure consisting of fluid-filled concentric lamellae.
- Central neurite ending responsible for mechanosensitive ion channel activation.

- **Functionality:**

- Acts as a biological **pass-band filter**.
- The fluid-lamellar system transmits dynamic stresses to the neurite while dampening static pressure.

- **Modeling Goal:**

- Correlate mechanical displacement with neural action potential frequency.

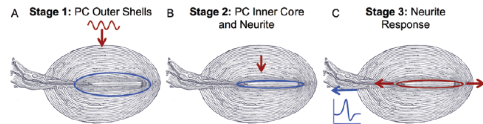


Figure: Pacinian corpuscle[1]

Introduction: Computational Pipeline

Methodology: Fully coupled multi-physics approach

Structural Mechanics

Viscoelastic modeling
& geometry optimization



Electrophysiology

General Form PDEs
(Hodgkin-Huxley dynamics)



Integrated Output

Tracking of mechano-electrical Action Potential propagation

Model Geometry: Neural Structure

Approach: 2D Axisymmetric Model to optimize computation.

- **Neurite:** Cylindrical.
- **Terminal Bulb:** Distal spherical ending.
- **Filopodia:** Three mechanical sensors to capture deformations:
 - 1 At half the length of the neurite.
 - 2 At the extreme distal point of the terminal bulb.
 - 3 On the bulb at an angle of 66° .

Component	Parameter	Value (μm)
Neurite	Radius (R_{neur})	1.5
Neurite	Length (L_{neur})	600
Terminal Bulb	Radius (R_{bulb})	5.0
Filopodia	Width (W_{filo})	1.6
Filopodia	Height (H_{filo})	0.6

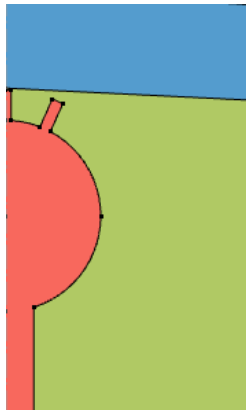


Figure: Bulb Ending

Model Geometry: Core and Lamellar Structure

- **Inner Core:** Central supporting structure.
- **Outer Core (Filter):** Lamellae and interlamellar fluid acting as a high-pass mechanical filter.
- **Thin Structures (Membrane):** Lamellae are modeled as *thin structures* using COMSOL's **Membrane** module, maximizing efficiency without meshing the thickness.
- **Variable Thickness:** Lamellar thickness follows a **Weibull function** derived from the literature[2]:

$$f(n) = t_{lmin} + (t_{lmax} - t_{lmin}) \cdot \left(1 - \exp \left(- \left(\frac{n}{\alpha \cdot N_{lam}} \right)^\beta \right) \right)$$

Component	Parameter	Value[1]
Inner Core	Semiaxis (R_{inner})	300 μm
Outer Core	Number of Lamellae (N_{lam})	30
Outer Core	Interlamellar fluid thickness (t_{flu})	5 μm
Outer Core	Lamellar thickness (t_{lam})	0.1 - 0.4 μm

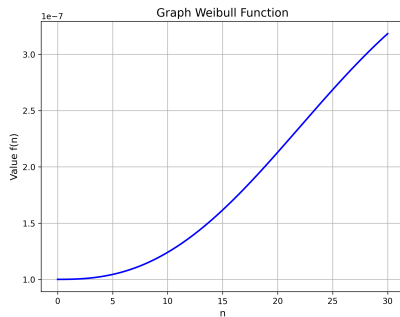


Figure: Weibull function modeling lamellar thickness variation.

Mechanics: Materials and Viscoelasticity

- **Viscoelasticity (Generalized Maxwell Model):** The interlamellar fluid is modeled with a generalized Maxwell model (a damper and a spring **in series**, in **parallel** with a spring)[3][4].
- **Solid Domains:** Neurite and Inner Core are modeled as nearly incompressible solid tissues.

Parameters from Literature[1]		
Domain	Parameter	Value
Neurite	Young's Modulus (E)	7 kPa
Inner Core	Young's Modulus (E)	700 kPa
Solid Tissues	Poisson's ratio (ν)	0.49
Tuning Parameters		
Domain	Parameter	Value
Interlamellar Fluid	Poisson's ratio (ν)	0.4
Interlamellar Fluid	Shear Modulus (G)	3 kPa
Interlamellar Fluid	Relaxation Time (τ)	0.07 ms

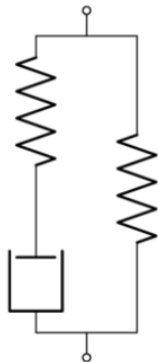


Figure: Generalized Maxwell model

Mechanics: Loading Conditions (Pressure Distribution)

Baseline Spatial Load: The baseline pressure field applied to the outer shell is defined as:

$$P(r, z) = -p_0 \cdot (C_{right} + \cos(2 \cdot \text{atan2}(r, z - R_{inner}))) , \quad \text{where } C_{right} = -\frac{\int_{outer} \cos(2\phi) dS}{\int_{outer} 1 dS}$$

- **Rationale:** Integral of the pressure over the surface is exactly zero. It imposes pure asymmetric "squeezing", **ignoring hydrostatic contributions**[1].
- **Temporal Modulations:** Baseline profile is adapted for specific studies:
 - **Freq. Sweep:** Used in frequency domain.
 - **Rectangular Load:** Multiplied by a rect function for sustained indentation.
 - **Sine Load:** Multiplied by $\sin(2\pi ft)$ for vibrotactile stimuli (e.g., 50/150 Hz). indentation.

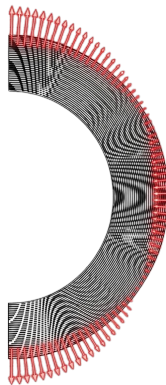


Figure: Applied load

Electric Model: Hodgkin-Huxley

Membrane Equation:

$$\frac{\partial V_m}{\partial t} = \frac{1}{C_m} \left[\frac{a\sigma_i}{2} \frac{\partial^2 V_m}{\partial x^2} - I_{ion} \right]$$

Total Ionic Current (I_{ion}):

$$I_{ion} = \bar{G}_{Na} m^3 h (V_m - V_{Na}) + \bar{G}_K n^4 (V_m - V_K) + G_L (V_m - V_L)$$

Gating Variables Kinetics:

$$\frac{dy}{dt} = \alpha_y(1 - y) - \beta_y y, \quad \text{for } y \in \{m, h, n\}$$

Initial Values

- **Resting Potential (V_r):** -60 mV
- **Gating Variables:**
 - $n = 0.3177$
 - $m = 0.0529$
 - $h = 0.5961$

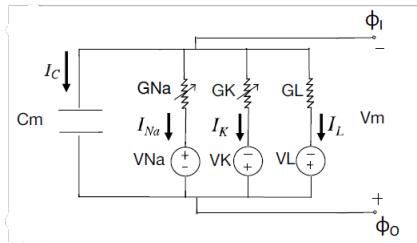


Figure: Hodgkin-Huxley Electric Circuit [5]

Electric Model: Mechano-Electrical Transduction

Core Mechanism: Translating mechanical strain into stimulating current via a sigmoidal activation function.

- **The Activation Function:**

- Based on the sigmoidal relationship proposed by Quindlen et al. (2016)[1] for rapidly relaxing currents.
- Computes the stimulating current (I_{stim}) triggered by the maximum first principal strain (ε_{max}) at the filopodia.

$$I_{stim} = \frac{I_{max}}{1 + \exp(-800 \cdot (\varepsilon_{max} - \text{base}))}$$

- **Parameters:**

- $I_{max} = 3 \text{ A/m}^2$
- $\text{base} = 0.011$ (1.1% strain threshold)

- **COMSOL Implementation:**

- Total Boundary Source (on each filopodia): $I_{total} = -I_{ion} + I_{stim}$

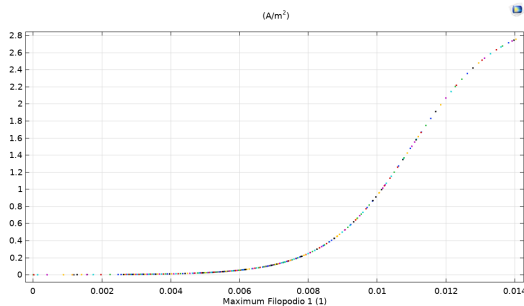


Figure: Stimulation Current vs. Filopodium Strain

Human Adaptation: Fast-Spiking Kinetics

Objective: Scaling the original HH model to replicate mammalian FA mechanoreceptor dynamics.

- **The Physiological Gap:**

- The standard equations were derived from the *Loligo* squid giant axon.
- Baseline kinetics are intrinsically too slow to replicate the high-frequency spiking of human sensory nerves.

- **Conductance Tuning:** Human neural membranes exhibit distinct ionic permeability profiles. We implemented an increased maximum sodium conductance ($\bar{G}_{Na} = 180 mS/cm^2$ [6]) to allow for rapid, sharp depolarizations.

- **Kinetic Acceleration:** To match human spike widths and frequencies, the transition rate constants (α, β) are multiplied by an experimentally derived scaling factor (k_{temp}) [7]:

$$\frac{dy}{dt} = k_{temp} [\alpha_y(V)(1 - y) - \beta_y(V)y], \quad y \in \{m, h, n\}$$

Study: Frequency Response

Objective: Evaluate the frequency-dependent mechanical response at the neurite filopodia.

- **Band-Pass Filter Behavior:**

- The model successfully replicates the physiological band-pass nature of the receptor.
- The Pacinian corpuscle detects vibrations in the 20-1000 Hz frequency range.

- **Filopodial Strain Amplification:**

- Maximum deformation is strongly localized at the filopodia.
- Regions of largest strain occur near the filopodia at the central and distal ends of the neurite.

- **Comparison with Literature:**

- Quindlen et al. (2016): peak sensitivity at 150 Hz.
- Our model: peak response in the 300-400 Hz range. This aligns perfectly with the most sensitive physiological vibrotactile range of the pacini, reported around $\approx 300\text{Hz}$ [8].

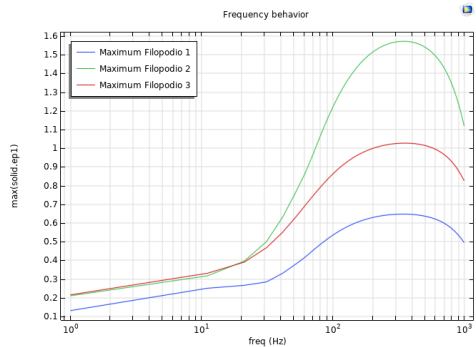


Figure: Maximum strain at filopodia vs. Frequency

Study: Frequency Response, Literature Comparison

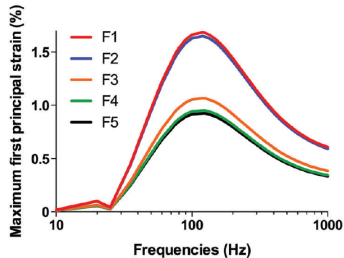


Figure: Literature (Quindlen et al., 2016)[1]

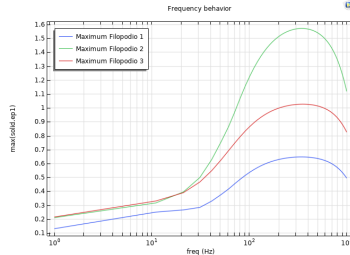


Figure: Unified Model

- The model confirms the band-pass nature of the Pacinian corpuscle, matching the filtering characteristics described in literature.
- **High-Frequency Shift:** Our peak sensitivity ($\approx 350\text{Hz}$) demonstrates a shift towards the optimal human vibrotactile range compared to the 150 Hz reported in previous works.
- **Methodological Validation:** The close alignment of results proves that our monolithic COMSOL integration captures the mechanical strain amplification of the lamellar structure as effectively as the decoupled multi-software approach.

Study: Rectangular Load, Mechanical Adaptation

The Rectangular Load Study: A sustained step indentation is applied to study the receptor's mechanical transient response.

$$P_{rect}(r, z, t) = A_{stim_rect} \cdot P(r, z) \cdot rect(t)$$

(Step function $rect(t)$ starts at 10 ms and ends at 50 ms)

Observations from the COMSOL Model:

- **Onset Peak ($t = 10\text{ ms}$):** The sudden application of the load translates into an immediate, sharp peak in the maximum principal strain (solid.ep1) inside the core.
- **Adaptation Phase (10 – 50 ms):** Despite the load being constantly sustained (plateau), the strain rapidly decays. This demonstrates the **Mechanical Adaptation** driven by the relaxation of the Maxwell fluid ($\tau = 0.07\text{ ms}$).
- **Offset Peak ($t = 50\text{ ms}$):** The removal of the load triggers a recoil, resulting in a second, distinct strain peak.

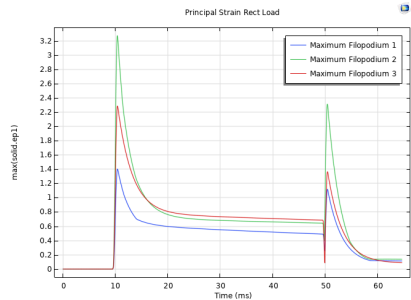


Figure: Adaptation of the maximum principal strain (solid.ep1).

Study: Rectangular Load, Electrical Response (Spikes)

From Strain to Action Potentials:

- **Activation Function:** Converts mechanical strain (solid.ep1) into an induced stimulation current .
- **Transient Output:** Strain peaks only at onset and offset, triggering the activation function exclusively during these phases.
- **FA-II Behavior:** The Hodgkin-Huxley equations generate exactly **two spikes** (push and release), replicating *Fast-Adapting* responses, this is the same results obtained in literature [1].

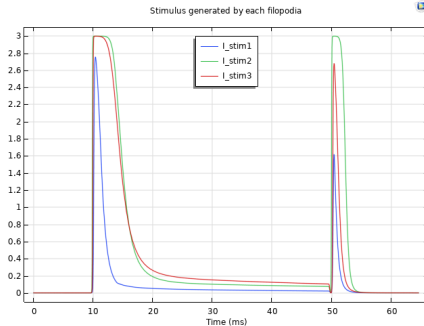


Figure: Activation Function (I_{stim}).

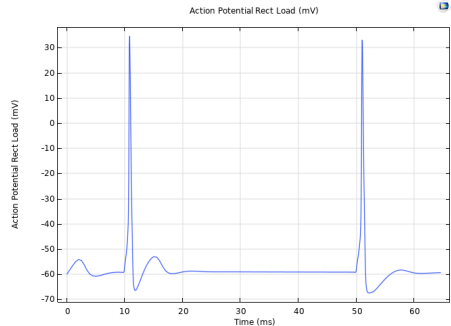


Figure: Generated Action Potentials.

Study: Sinusoidal Load, Mechanical Frequency Response

The Sinusoidal Load Study: A sinusoidal load applied at two frequencies ($f = 50 \text{ Hz}$ and 150 Hz):

$$P_{\text{sine}}(r, z, t) = A_{\text{stim_sine}} \cdot P(r, z) \cdot \sin(\omega t), \text{ where } \omega = 2\pi f$$

High-Pass Mechanical Filtering:

- **Graph Analysis:** the strain reaches significantly lower peak values at 50 Hz ($\text{max} \approx 1.1$) compared to 150 Hz ($\text{max} > 1.5$) for the exact same input amplitude.

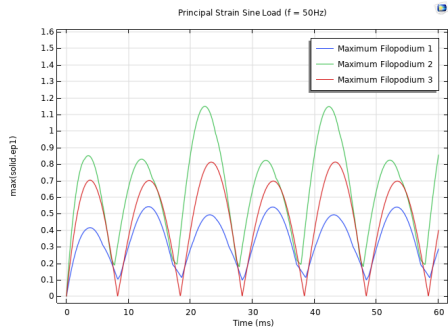


Figure: Filopodia Strain at 50 Hz .

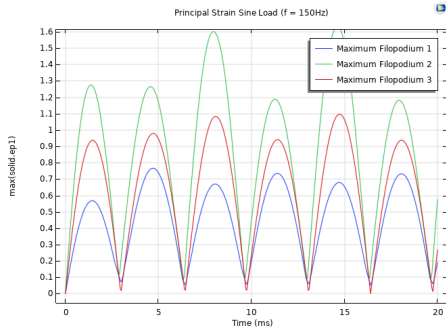


Figure: Filopodia Strain at 150 Hz .

Study: Sinusoidal Load, Electrical Response (Spikes)

Spiking Behavior and Frequency Sensitivity:

- **Temporal Windowing:** Simulation time for the 50 Hz load is **three times longer** than at 150 Hz to ensure a fair comparison of spikes **per stimulus cycle**.
- **Frequency-Dependent Sensitivity:** At constant amplitude, higher frequencies (150 Hz) elicit spikes, while lower frequencies (50 Hz) fail. As per literature [1], low frequencies require higher amplitudes to trigger action potentials.

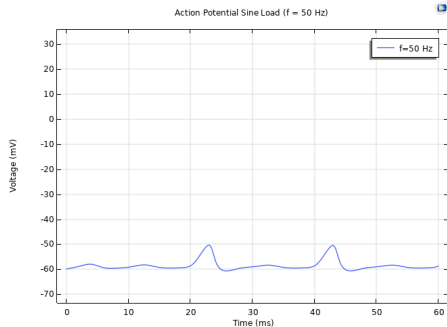


Figure: Action Potentials at 50 Hz

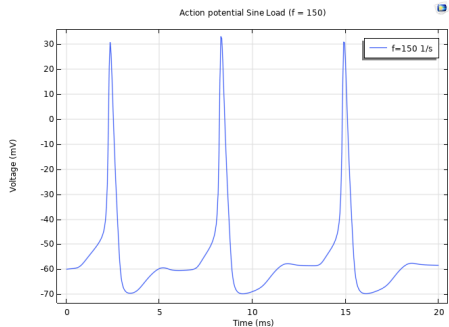


Figure: Action Potentials at 150 Hz

Conclusions and Future Work

Conclusions

- **Multiphysics Coupling:** Successfully integrated the viscoelastic mechanical response of the lamellae with the electrophysiological neural model.
- **Frequency Sensitivity:** Validated the receptor's pass-band filter behavior. Higher frequencies readily elicit action potentials, while lower frequencies require higher stimulus amplitudes.
- **Receptor Adaptation:** The model accurately captures the fast-adapting nature of the Pacinian corpuscle under rectangular static loads.

Future Work

- **3D Geometry & Realistic Stimuli:** Transition to a full 3D anatomical model and test the system against stochastic or complex tactile textures.
- **Bioinspired Tactile Sensors:** Leverage the model's insights to design novel artificial mechanoreceptors that mimic the Pacinian architecture.
- **Morphological Computation:** Exploit the viscoelastic lamellae as a physical processing layer to filter and pre-process mechanical stimuli, significantly offloading the electronic computational effort.

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Thank You!

Any Questions?